



REDROB · DATA & AI CHALLENGE

HireFit Ranker

Ranks **careers**, not **keywords**.

A deterministic, offline, CPU-only engine that reads who actually fits a role — then returns a 100-candidate shortlist a recruiter can trust.

100K

candidates scored

80s

cloud CPU runtime (≤ 300 s)

0.894

NDCG@10 *

0

honeypots in top-100

* NDCG@10 on an independent LLM-judged sample (dev proxy) — see slide 9. Not the hidden challenge score.

HIREFIT RANKER

THE PROBLEM

Keyword filters miss the right people — and rank the wrong ones first.

The dataset is engineered to punish keyword matching. Four trap types sink a naive ranker.



Keyword stuffers

Wrong-role profiles padded with AI skills — a Marketing Manager listing 9 AI keywords.



Wrong-role profiles

Perfect skill lists attached to titles that never shipped a ranking system.



Honeypots

Subtly impossible profiles — “expert” in 10 skills with 0 months used. Forced to tier 0.



Unavailable-but-perfect

Great on paper, but 5% recruiter response and not logged in for months.



The real task: rank candidates the way a great recruiter would — reading career history and behavior, not counting keywords. A candidate who *built* a recommender at a product company outranks one who merely *lists* “RAG” and “Pinecone.”

The JD lists keywords; the role is hiring for evidence.

Our whole design follows from reading the gap between what the JD says and what it means.



What the JD says

“5–9 years, embeddings, retrieval, ranking, NLP, deep learning, PyTorch, TensorFlow...”

embeddings

retrieval

ranking

NLP

5–9 yrs

The trap: skill-keyword overlap is high for stuffers and low for the engineers who actually shipped the systems.



What the role means



Shipped work — Built & deployed ranking / search / recsys in production.



Career trajectory — Consistent growth in relevant, product-company roles.



Right seniority — anchored on the JD's 5–9 band — juniors demoted hard; senior depth read from trajectory, not years arithmetic.



Actually available — Responsive, open to work, reachable notice period.

We measured the config that scores higher — and declined it on principle.

Every knob swept under a decision rule committed before the first run; the JD is the spec, not the judge.

0.896

SHIPPED (LLM-JUDGE)

0.926

FLOOR-OFF — BIASED
SAMPLE, DECLINED

0

RANK CHANGES 0.25–0.55

11 / 11

PLANTED TRAPS CAUGHT



How the rule decided

Rule first committed before any sweep run — ties break toward the JD's down-weight instruction

Coverage guard floor 0.85–1.00 configs excluded: <95% label coverage, selection-biased

Winner shipped config — identical top-100 across floors 0.25–0.55



We built our strongest rival — it lost

A LambdaMART challenger built on our own 33 features plus recovered generator structure lost its pre-registered gate: 0.900 vs 0.906. One of four measured negatives — embeddings, learned-LR, the challenger, and a declined availability hedge. The adopted change is role-family depth scoring; an 8-swap calibration pass was trialed then removed — the official path now uses NO candidate-ID tuning (CI-enforced).



Honest framing: the pool plants 11 perfect-on-paper-but-unavailable gold profiles — one a honeypot in gold clothing — the guardrails demote all 11; an LLM judge missed one of them.

THE SYSTEM

Deterministic at rank time — every choice gated by LLM judges offline.

Six CPU-only stages, no network or GPU; tuned and frozen under two LLM judges and a LambdaMART challenger gate.



Final score = base score × behavioral × honeypot × disqualifier

Guardrails are multiplicative — a strong-on-paper profile that is impossible, off-target, or unreachable is pushed down no matter how good its keywords look.

RUNS UNDER:

✓ Offline

✓ CPU-only

✓ No GPU

✓ Deterministic

✓ bm25s backend

33 features separate keyword lists from recruiter-plausible fit.

Each signal is computed deterministically from profile facts — and every rank decomposes back into them.



Skills & depth

10 features

core / nice-to-have match, skill depth, endorsement trust, assessment scores, GitHub, semantic (HyRE) match



Career evidence

11 features

product-company ratio, IR/ranking experience, production evidence, seniority, trajectory, scale, role-family depth (backend, data/BI)



Experience fit

4 features

years-of-experience band, education, ML/AI tenure, open-source signal



Behavior

6 features

availability, engagement, responsiveness, interview reliability, profile quality, notice



Logistics

2 features

preferred-location match, willingness to relocate



Multiplicative guardrails

Behavioral, honeypot & disqualifier multipliers gate the base score so traps can't buy their way to the top.

The shortlist is real: every rank is traceable.

Live output from the released 100K pool, plus the exact feature values and guardrails behind the #1 candidate.

Ranked output

submission.csv top 6

Rank	Candidate evidence	YOE	Score
1	Senior ML Engineer - Zomato hybrid retrieval + ranking at 50M+ queries	7.2	1.000
2	Senior NLP Engineer - Niramai semantic search + embeddings + strong response	7.8	0.940
3	Senior ML Engineer - Genpact AI IR, pgvector, Sentence Transformers	6.1	0.933
4	Senior AI Engineer - Apple FAISS + fine-tuning + 30-day notice	5.9	0.931
5	Senior ML Engineer - Flipkart OpenSearch + LLM fine-tuning; logistics flag	8.1	0.921
6	Senior AI Engineer - Netflix Elasticsearch + Qdrant + PEFT	7.8	0.902

Readout: the top six are senior AI/ML/NLP/search profiles with production evidence; tie-breaks are deterministic by candidate_id.

Why #1 wins

CAND_0018499

Senior ML Engineer at Zomato | 7.2 yrs | Noida | 15-day notice

core_skill_match

0.982

production_evidence

1.000

ir_ranking_experience

1.000

product_company_ratio

1.000

career_trajectory_score

1.000

behavior x1.10

honeypot x1.00

DQ x1.00

Grounded reason: shipped retrieval/ranking; skills include Learning to Rank + Embeddings; response 0.61.

Stuffers and honeypots never reach the shortlist.

Measured on the released pool — the guardrails do real work, not decoration.



Keyword stuffers buried

4,800+

Wrong-role profiles across the pool's 12 non-target strata — zero reach the top-100.



Honeypots blocked

53 → 0

Impossible profiles detected, then zero emitted in the top 100. Challenge DQ threshold is >10%.



Unavailable down-weighted

×0.42

Behavioral multiplier on an 8-yr profile with 12% response & not-open-to-work — pushed off the shortlist.



Plain-language fit is captured too: skill matching runs over the full profile text, not just the skills array — so an engineer who *describes* building a recommender (without the buzzwords) still scores on core fit.

VALIDATION

Three independent LLM judge families confirm the top-10 — it isn't self-graded.

We avoided grading our ranker with its own heuristics. A separate judge scored a stratified sample.

0.894

NDCG@10

0.873

NDCG@50

0.912

MAP

1.00

P@10



How it was judged

249 stratified candidates — ranker top ranks + higher heuristic tiers + a random control

Judges gemini-2.5-flash + gpt-4.1-mini + deepseek, same rubric, same 249 ids: kappa 0.918–0.935 across all pairs.

Top-10 all 4–5 (judge 1); all tier-5, NDCG@10 = 1.0 (judge 2)



Not propped up by our own labels

The judge composite (0.896) is slightly higher than our non-circular heuristic harness (0.886) — the ranking is strong by an outside opinion, not just our own.



Honest framing: a development-time proxy on a 249-candidate sample, not the hidden challenge score. An 8-swap calibration pass was trialed (judge re-test +0.0124) but later removed; the shipped ranking uses no candidate-ID tuning (docs/llm_judge_eval_3.md).

Reproducibility beats raw model power.

The deck does not hide limitations; it turns them into Stage-5 defense points judges can verify.

Defensible decisions

shipped path

- **No LLM per candidate**
100K hosted calls break the CPU/no-network budget and make reproduction fragile.
- **Dense embeddings tested, then kept off**
NDCG@10 was flat while runtime and complexity rose, so the simpler ranker won.
- **Guardrails are multiplicative**
Impossible, off-target, or unreachable profiles cannot buy rank with keywords.
- **Deterministic reasoning**
Same input gives the same CSV; every sentence points to real profile facts.

Next hardening moves

honest roadmap

- **Hidden ground truth**
The official leaderboard is unseen, so proxy eval is evidence - not a score claim.
- **Conservative hard-trap recall**
53 hard traps are blocked; softer suspicious profiles are down-weighted, not over-zeroed.
- **Learned ranking layer**
Future labels could train LambdaMART/XGBoost over these same features.
- **Stage-3 fallback**
If multiprocessing is constrained, --workers 1 preserves the exact same ranking output.

REPRODUCIBILITY

One command reproduces the shortlist — offline, well under the five-minute limit.

Built for the Stage-3 sandbox: it runs unmodified inside the challenge's compute box.

```
>_ python rank.py --candidates ./candidates.jsonl --out ./submission.csv
```

CONSTRAINT

CHALLENGE LIMIT

MEASURED

Runtime

≤ 5 min (300s)

80–125 s



Memory

≤ 16 GB RAM

~6.1 GB peak



Compute

CPU only

CPU only



Network

off during ranking

fully offline



Submission package

- ✓ GitHub repo — clean, tested, reproducible
- ✓ submission.csv — top-100 + grounded reasoning
- ✓ METHODOLOGY.md — every choice, with the why
- ✓ Live sandbox — HuggingFace Space
- ✓ Independent eval evidence committed (249 labels)

A drop-in scoring layer for Redrob's hiring stack — auditable by design.

Ranks talent on behavior and evidence, not resumes — the same thesis as the platform it would join.

~3 ms

AUDIT API, MEASURED

1 cmd

FULL REPRODUCTION

0

LLM CALLS PER RANKED
CANDIDATE

33

EXPLAINABLE FEATURES



Integration path

JD compiler any role text compiles into the same deterministic scoring program

Audit payload per-candidate JSON: features, multipliers, flags — recruiter-readable

Signals behavioral multipliers consume Redrob activity data natively



Roadmap, honestly

Indic-script normalization is a localized swap (one module) — the architecture is script-agnostic. When real outcome labels exist, LambdaMART can replace the hand weights over the same 28 features without touching the audit trail: the features are the durable asset.



LLMs where they are strong — offline judging and evaluation — and deterministic evidence where hiring decisions need to be defended.

THE TAKEAWAY

We don't rank resumes. We rank **hireability**.



Reads careers & behavior, not buzzwords



Every rank explained by 33 named features



Top-10 verified by an independent judge



Blocks honeypots, stuffers & the unavailable



100K scored offline, on CPU, in 80s on cloud hardware